

CSE-4632

Digital Signal Processing Lab

Lab 4

Lab Group: **1**

Thursday, 8th July, 2026

Task 1

Build and Verify a Sinusoid Generator

Implement the function below and use it to generate and plot signals.

```python
import numpy as np
import matplotlib.pyplot as plt

def make_sinusoid(kind, A, f, phi, t):
    """
    kind : 'sin' or 'cos'
    A    : amplitude
    f    : frequency in Hz
    phi  : phase offset in radians
    t    : numpy array of time samples
    returns: numpy array x(t)
    """
    # TODO: implement
    pass
```

Build:

(a) Implement `make_sinusoid`.

(b) Generate $t = 1000$ points over $[0, 2]$ seconds. Plot `make_sinusoid('sin', 1, 1, 0, t)` and `make_sinusoid('cos', 1, 1, 0, t)` on the same axes.

(c) Write an assertion-based test (no plotting) that checks `make_sinusoid('cos', 1, f, 0, t)` equals `make_sinusoid('sin', 1, f, np.pi/2, t)` to within `1e-9` for three different frequencies of your choice. Use `np.allclose`.

(d) Extend your script to synthesise a signal from a sum of sinusoids:

$$x(t) = 2\sin(2\pi \cdot 3t) + 1\cos(2\pi \cdot 7t + 0.5)$$

Plot it over $[0, 1]$ seconds and print its peak-to-peak amplitude using `np.ptp`.

Task 2: Forward DFT From Scratch

Implement the Analysis Equations

```python
import numpy as np

def my_dft(x):
    """
    x : real-valued 1D numpy array, length N
    returns: ReX, ImX, each of length N//2 + 1
    Implement the analysis equations directly with nested loops --
    do not use np.fft.
    """
    N = len(x)
    ReX = np.zeros(N // 2 + 1)
    ImX = np.zeros(N // 2 + 1)
    # TODO: implement the double loop over k and n
    return ReX, ImX
```

Build:

(a) Complete `my_dft`.

(b) Run it on `x = np.array([1, 2, 3, 4])` and print `ReX`, `ImX`.

(c) Write an assertion test that checks your output against the known result $X[0] = (10, 0)$, $X[1] = (-2, 2)$, $X[2] = (-2, 0)$ using `np.allclose`.

(d) Write a second function `magnitude_spectrum(ReX, ImX)` returning $\sqrt{\mathrm{Re}X^2 + \mathrm{Im}X^2}$, and plot it as a stem plot for a 32-sample signal $x[n] = \sin(2\pi \cdot 4n/32) + 0.5\cos(2\pi \cdot 9n/32)$. Confirm the two expected peaks appear at $k = 4$ and $k = 9$.

Task 3: Inverse DFT and Round-Trip Reconstruction

Implement the IDFT and Measure Reconstruction Error

```python
import numpy as np

def my_idft(ReX, ImX, N):
    """
    ReX, ImX : half-spectrum arrays of length N//2 + 1
    N        : original signal length
    returns  : reconstructed real signal x[n], length N
    Implement the synthesis equation with the correct K_k scaling
    for interior bins vs DC/Nyquist bins.
    """
    x = np.zeros(N)
    # TODO: implement
    return x
```

Build:

(a) Implement `my_idft`.

(b) Round-trip test: for $x[n] = \{1, 2, 3, 4\}$, compute `ReX, ImX = my_dft(x)` then `x_rec = my_idft(ReX, ImX, 4)`. Assert `np.allclose(x, x_rec)`.

(c) Repeat the round-trip test for five random real signals of varying even lengths ($N = 8, 16, 32, 64, 128$) and report the maximum absolute reconstruction error for each, using `np.max(np.abs(x - x_rec))`.